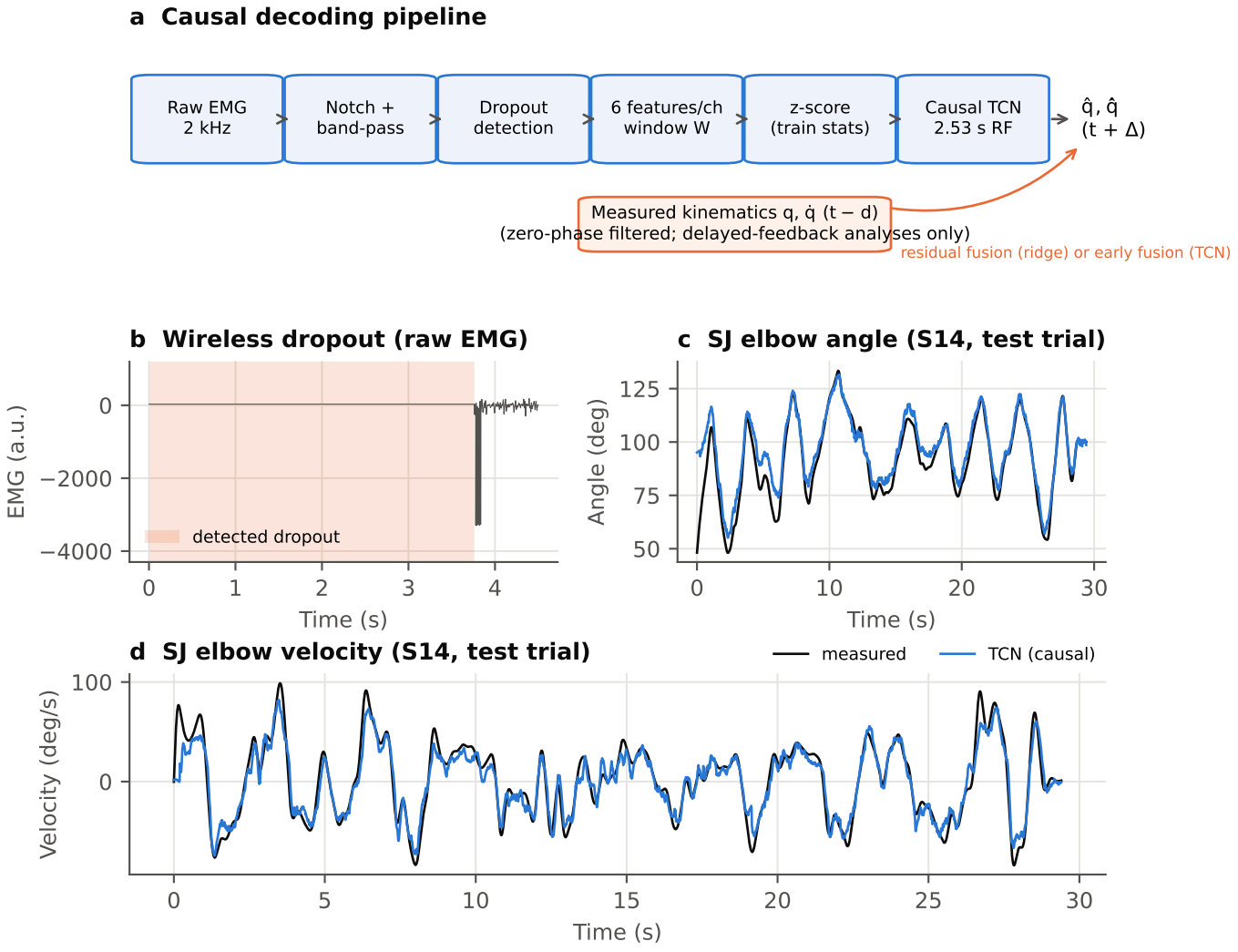

a Causal decoding pipeline
Raw EMG 2 kHz
Notch + band-pass
Dropout detection
6 features/ch window W
z-score (train stats)
Causal TCN 2.53 s RF
q̂, q̂̇ (t + Δ)
Measured kinematics q, q̇ (t − d)
(zero-phase filtered; delayed-feedback analyses only)
residual fusion (ridge) or early fusion (TCN)
b Wireless dropout (raw EMG)
EMG (a.u.)
detected dropout
Time (s)
c SJ elbow angle (S14, test trial)
Angle (deg)
Time (s)
d SJ elbow velocity (S14, test trial)
measured
TCN (causal)
Velocity (deg/s)
Time (s)